Exercícios Programados 5 - Gabarito

1^a **Questão:** Esboce a região do plano complexo que corresponde a:

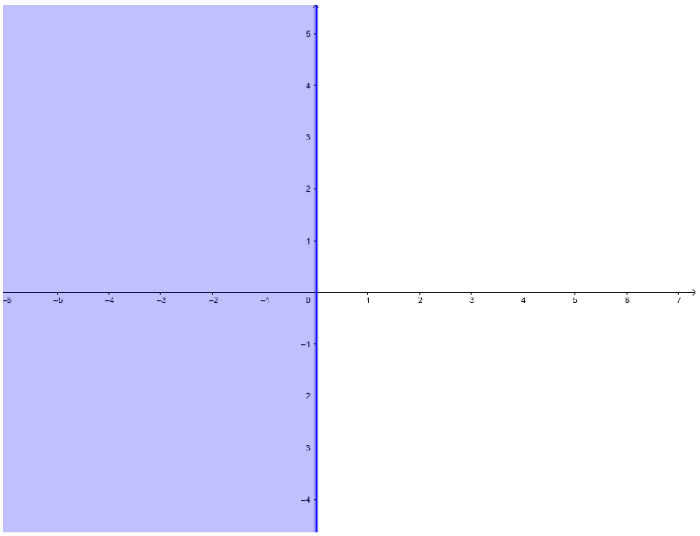
$$\textbf{(a)}\ \left|\frac{z+1}{z-1}\right| \leq 1 \qquad \textbf{(b)}\ Re\left(\frac{z+1}{z-1}\right) = 0$$

Solução: Consideremos $z = x + yi$

(a)

$$\left|\frac{z+1}{z-1}\right| \leq 1 \iff |z+1| \leq |z-1| \iff |z+1|^2 \leq |z-1|^2 \iff |(x+1)+yi|^2 \leq |(x-1)+yi|$$

$$\iff (x+1)^2 + y^2 \leq (x-1)^2 + y^2 \iff 2x \leq -2x \iff 4x \leq 0 \iff x \leq 0$$

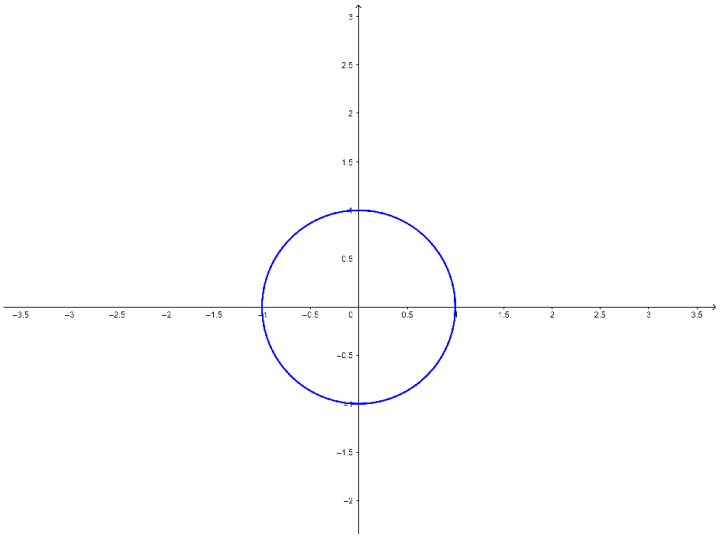


(b) Note que:

$$\frac{z+1}{z-1} = (z+1)\frac{\overline{z-1}}{|z-1|^2} = \frac{(z+1)(\overline{z}-1)}{|z-1|^2} = \frac{z\overline{z} - z + \overline{z} - 1}{|z-1|^2} = \frac{1}{|z-1|^2}\left(x^2 + y^2 - 1 - 2yi\right)$$

Dessa forma,

$$Re\left(\frac{z+1}{z-1}\right) = 0 \iff x^2 + y^2 - 1 = 0 \iff x^2 + y^2 = 1$$



2^a **Questão:** Para os complexos abaixo z_1 e z_2, determine a representação polar de $z_1 z_2$ e $\dfrac{z_1}{z_2}$ de duas formas diferentes:

- Obter primeiro as formas polares de z_1 e z_2 e depois efetuar as operações na forma polar.
- Efetuar primeiro as operações na forma algébrica e depois obter a forma polar dos resultados correspondentes.

(a) $z_1 = 1 + \sqrt{3}i$ e $z_2 = -\sqrt{3} + i$

(b) $z_1 = -2 + 2i$ e $z_2 = -\dfrac{1}{8} - \dfrac{1}{8}i$

Solução:

(a)

Primeira Forma:

$$\rho_1 = \sqrt{1^2 + \left(\sqrt{3}\right)^2} = \sqrt{4} = 2$$

$$\begin{cases} \cos(\theta_1) = \dfrac{Re(z_1)}{\rho_1} = \dfrac{1}{2} \\ \sin(\theta_1) = \dfrac{Im(z_1)}{\rho_1} = \dfrac{\sqrt{3}}{2} \end{cases} \Longrightarrow \theta_1 = \frac{\pi}{3} \Longrightarrow z_1 = 2\left(\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)\right)$$

$$\rho_2 = \sqrt{\left(-\sqrt{3}\right)^2 + 1^2} = \sqrt{4} = 2$$

$$\begin{cases} \cos(\theta_2) = \dfrac{Re(z_2)}{\rho_2} = \dfrac{-\sqrt{3}}{2} \\ \sin(\theta_2) = \dfrac{Im(z_2)}{\rho_2} = \dfrac{1}{2} \end{cases} \implies \theta_2 = \frac{5\pi}{6} \implies z_2 = 2\left(\cos\left(\frac{5\pi}{6}\right) + i\sin\left(\frac{5\pi}{6}\right)\right)$$

Dessa forma, segue que

$$z_1 z_2 = 2 \cdot 2\left(\cos\left(\frac{\pi}{3} + \frac{5\pi}{6}\right) + i\sin\left(\frac{\pi}{3} + \frac{5\pi}{6}\right)\right) = 4\left(\cos\left(\frac{7\pi}{6}\right) + i\sin\left(\frac{7\pi}{6}\right)\right)$$

e

$$\frac{z_1}{z_2} = \frac{2}{2}\left(\cos\left(\frac{\pi}{3} - \frac{5\pi}{6}\right) + i\sin\left(\frac{\pi}{3} - \frac{5\pi}{6}\right)\right) = \cos\left(\frac{-\pi}{2}\right) + i\sin\left(\frac{-\pi}{2}\right)$$

Segunda Forma:

$$z_1 z_2 = (1 \cdot (-\sqrt{3}) - \sqrt{3} \cdot 1) + (1 \cdot 1 + \sqrt{3} \cdot (-\sqrt{3}))i = -2\sqrt{3} - 2i$$

$$\rho = \sqrt{\left(-2\sqrt{3}\right)^2 + (-2)^2} = \sqrt{16} = 4$$

$$\begin{cases} \cos(\theta) = \dfrac{Re(z_1 z_2)}{\rho} = \dfrac{-\sqrt{3}}{2} \\ \sin(\theta) = \dfrac{Im(z_1 z_2)}{\rho} = \dfrac{-1}{2} \end{cases} \implies \theta = \frac{7\pi}{6} \implies z_1 z_2 = 4\left(\cos\left(\frac{7\pi}{6}\right) + i\sin\left(\frac{7\pi}{6}\right)\right)$$

$$\begin{aligned} \frac{z_1}{z_2} &= z_1 \frac{\overline{z_2}}{|z_2|^2} = (1 + \sqrt{3}i)\left(\frac{-\sqrt{3}}{4} - \frac{1}{4}i\right) \\ &= \left(1 \cdot \left(\frac{-\sqrt{3}}{4}\right) - \sqrt{3} \cdot \left(\frac{-1}{4}\right)\right) + \left(1 \cdot \left(\frac{-1}{4}\right) + \sqrt{3} \cdot \left(\frac{-\sqrt{3}}{4}\right)\right) i = -i \end{aligned}$$

$$\rho = \sqrt{0^2 + (-1)^2} = 1$$

$$\begin{cases} \cos(\theta) = \dfrac{Re(z_1/z_2)}{\rho} = \dfrac{0}{1} = 0 \\ \sin(\theta) = \dfrac{Im(z_1/z_2)}{\rho} = \dfrac{-1}{1} = -1 \end{cases} \implies \theta = \frac{3\pi}{2} \implies \begin{aligned} \frac{z_1}{z_2} &= \cos\left(\frac{3\pi}{2}\right) + i\sin\left(\frac{3\pi}{2}\right) \\ &= \cos\left(\frac{-\pi}{2}\right) + i\sin\left(\frac{-\pi}{2}\right) \end{aligned}$$

(b)

Primeira Forma:

$$\rho_1 = \sqrt{(-2)^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$$

$$\begin{cases} \cos(\theta_1) = \dfrac{Re(z_1)}{\rho_1} = \dfrac{-2}{2\sqrt{2}} = \dfrac{-\sqrt{2}}{2} \\ \sin(\theta_1) = \dfrac{Im(z_1)}{\rho_1} = \dfrac{2}{2\sqrt{2}} = \dfrac{\sqrt{2}}{2} \end{cases} \implies \theta_1 = \frac{3\pi}{4} \implies z_1 = 2\sqrt{2}\left(\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)\right)$$

$$\rho_2 = \sqrt{\left(-\frac{1}{8}\right)^2 + \left(-\frac{1}{8}\right)^2} = \sqrt{\frac{1}{32}} = \frac{1}{4\sqrt{2}}$$

$$\begin{cases} \cos(\theta_2) = \dfrac{Re(z_2)}{\rho_2} = \dfrac{-\sqrt{2}}{2} \\ \sin(\theta_2) = \dfrac{Im(z_2)}{\rho_2} = \dfrac{-\sqrt{2}}{2} \end{cases} \implies \theta_2 = \frac{5\pi}{4} \implies z_2 = \frac{1}{4\sqrt{2}}\left(\cos\left(\frac{5\pi}{4}\right) + i\sin\left(\frac{5\pi}{4}\right)\right)$$

Dessa forma, segue que

$$z_1 z_2 = 2\sqrt{2} \cdot \frac{1}{4\sqrt{2}}\left(\cos\left(\frac{3\pi}{4} + \frac{5\pi}{4}\right) + i\sin\left(\frac{3\pi}{4} + \frac{5\pi}{4}\right)\right) = \frac{1}{2}\left(\cos(2\pi) + i\sin(2\pi)\right)$$

e

$$\frac{z_1}{z_2} = \frac{2\sqrt{2}}{\frac{1}{4\sqrt{2}}}\left(\cos\left(\frac{3\pi}{4} - \frac{5\pi}{4}\right) + i\sin\left(\frac{3\pi}{4} - \frac{5\pi}{4}\right)\right) = 16\left(\cos\left(\frac{-\pi}{2}\right) + i\sin\left(\frac{-\pi}{2}\right)\right)$$

Segunda Forma:

$$z_1 z_2 = \left(-2 \cdot \left(-\frac{1}{8}\right) - 2 \cdot \left(-\frac{1}{8}\right)\right) + \left(-2 \cdot \left(-\frac{1}{8}\right) + 2 \cdot \left(-\frac{1}{8}\right)\right) i = \frac{1}{2}$$

$$\rho = \sqrt{\left(\frac{1}{2}\right)^2 + 0^2} = \frac{1}{2}$$

$$\begin{cases} \cos(\theta) = \dfrac{Re(z_1 z_2)}{\rho} = 1 \\ \sin(\theta) = \dfrac{Im(z_1 z_2)}{\rho} = 0 \end{cases} \implies \theta = 0 \implies \begin{aligned} z_1 z_2 &= \frac{1}{2}\left(\cos(0) + i\sin(0)\right) \\ &= \frac{1}{2}\left(\cos(2\pi) + i\sin(2\pi)\right) \end{aligned}$$

$$\frac{z_1}{z_2} = z_1 \frac{\overline{z_2}}{|z_2|^2} = (-2 + 2i)(-4 + 4i) = ((-2) \cdot (-4) - 2 \cdot 4) + ((-2) \cdot 4 + 2 \cdot (-4))i = -16i$$

$$\rho = \sqrt{0^2 + (-16)^2} = 16$$

$$\begin{cases} \cos(\theta) = \dfrac{Re(z_1/z_2)}{\rho} = \dfrac{0}{16} = 0 \\ \sin(\theta) = \dfrac{Im(z_1/z_2)}{\rho} = \dfrac{-16}{16} = -1 \end{cases} \Longrightarrow \theta = \frac{3\pi}{2} \Longrightarrow \begin{array}{rl} \dfrac{z_1}{z_2} & = 16\left(\cos\left(\dfrac{3\pi}{2}\right) + i\sin\left(\dfrac{3\pi}{2}\right)\right) \\ & = 16\left(\cos\left(\dfrac{-\pi}{2}\right) + i\sin\left(\dfrac{-\pi}{2}\right)\right) \end{array}$$

3^a **Questão:** Que valores pode ter $i^m + i^{-m}$, $m \in \mathbb{Z}$?

Solução: Escrevndo i na Forma Polar, temos:

$$i = \cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right)$$

- $m = 0 \Longrightarrow i^0 + i^{-0} = 2i^0 = 2\left(\cos\left(0 \cdot \dfrac{\pi}{2}\right) + i\sin\left(0 \cdot \dfrac{\pi}{2}\right)\right) = 2$
- $m \neq 0$

$$\begin{aligned} i^m + i^{-m} &= \cos\left(\frac{m\pi}{2}\right) + i\sin\left(\frac{m\pi}{2}\right) + \cos\left(\frac{-m\pi}{2}\right) + i\sin\left(\frac{-m\pi}{2}\right) \\ &= \cos\left(\frac{m\pi}{2}\right) + i\sin\left(\frac{m\pi}{2}\right) + \cos\left(\frac{m\pi}{2}\right) - i\sin\left(\frac{m\pi}{2}\right) = 2\cos\left(\frac{m\pi}{2}\right) \end{aligned}$$

Note que:

$$2\cos\left(\frac{m\pi}{2}\right) = \begin{cases} 0, \text{ se } m \text{ é ímpar} \\ 2, \text{ se } m \text{ é par e múltiplo de } 4 \\ -2, \text{ se } m \text{ é par e não é múltiplo de } 4 \end{cases}$$

Conclusão: Os possíveis valores para $i^m + i^{-m}$ são: 0, 2 e -2.

4^a **Questão:** Calcule $\left(\dfrac{1}{2} + \dfrac{\sqrt{3}}{2}i\right)^3$

Sugestão: Utilize a Forma Polar

Solução: Inicilamente, vamos escrever $z = \dfrac{1}{2} + \dfrac{\sqrt{3}}{2}i$ na forma polar. Note que

$$\rho = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1$$

e

$$\begin{cases} \cos(\theta) = \dfrac{Re(z)}{\rho} = \dfrac{1}{2} \\ \sin(\theta) = \dfrac{Im(z)}{\rho} = \dfrac{\sqrt{3}}{2} \end{cases} \Longrightarrow \theta = \frac{\pi}{3}.$$

Logo

$$z = \cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)$$

e, porntato, pela Fórmula de De Moivre, segue que:

$$z^6 = \cos\left(\frac{6\pi}{3}\right) + i\sin\left(\frac{6\pi}{3}\right) = \cos(2\pi) + i\sin(2\pi) = 1.$$

5^a **Questão:** Prove que

$$\begin{cases} \cos(3\theta) = \cos^3(\theta) - 3\cos(\theta)\sin^2(\theta) \\ \sin(3\theta) = -\sin^3(\theta) + 3\cos^2(\theta)\sin(\theta) \end{cases}$$

Sugestão: Desenvolva $(\cos(\theta) + i\sin(\theta))^3$ pelo Binômio de Newton e pela Fórmula de De Moivre.

Solução:

Pela Fórmula de De Moivre, temos:

$$(\cos(\theta) + i\sin(\theta))^3 = \cos(3\theta) + i\sin(3\theta).$$

Por outro lado, utilizando o desenvolvimento pelo Binômio de Newton, obtemos:

$$\begin{aligned} (\cos(\theta) + i\sin(\theta))^3 &= \cos^3(\theta) + 3\cos^2(\theta)i\sin(\theta) + 3\cos(\theta)(i\sin(\theta))^2 + (i\sin(\theta))^3 \\ . &= \left(\cos^3(\theta) - 3\cos(\theta)\sin^2(\theta)\right) + \left(-\sin^3(\theta) + 3\cos^2(\theta)\sin(\theta)\right)i \end{aligned}$$

Dessa forma, comparando as duas expressões, concluímos que

$$\begin{cases} \cos(3\theta) = \cos^3(\theta) - 3\cos(\theta)\sin^2(\theta) \\ \sin(3\theta) = -\sin^3(\theta) + 3\cos^2(\theta)\sin(\theta) \end{cases}$$